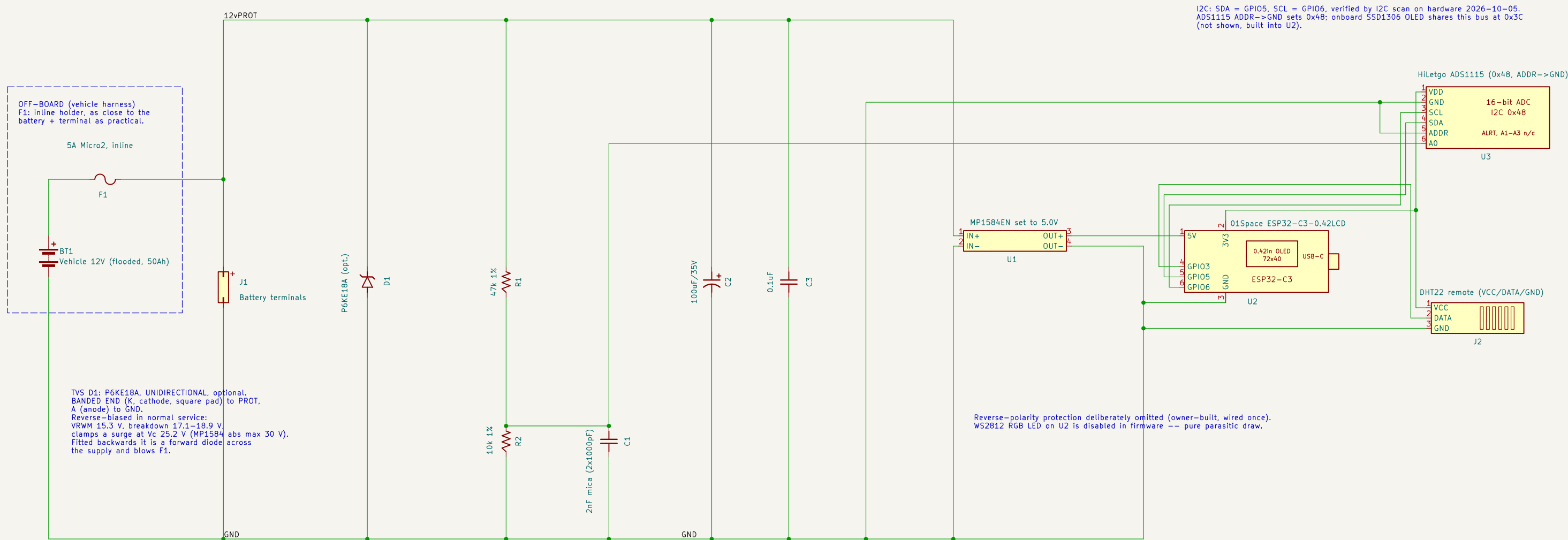


EV 12V BATTERY MONITOR
Battery -> WiFi -> HomeAssistant



```
I2C: SDA = GPIO5, SCL = GPIO6, verified by I2C scan on hardware 2026-10-05.
ADS1115 ADDR->GND sets 0x48; onboard SSD1306 OLED shares this bus at 0x3C
(not shown, built into U2).
```

Reverse-polarity protection deliberately omitted (owner-built, wired once).
WS2812 RGB LED on U2 is disabled in firmware -- pure parasitic draw.

J2: DHT22/AM2302, 1=VCC 2=DATA 3=GND. Data pull-up is on the sensor module.
For now the sensor is INSIDE the enclosure, hot-glued to the wall (reads a few degC warm, which errs safe). Remote option: cable out through a gland, max ~1 m at 3.3 V.

Divider R1 47k / R2 10k = 5.7:1 (unit 1 measured 46.5k/9.85k = 5.72:1). C1 = 2 x 1000pF mica in parallel (2nF, low leakage). ADS1115 abs max = VDD+0.3 = 3.6V.
12.22V(flooded 50% SOC)->2.14V 14.4V(charging)->2.52V 16.0V(design max)->2.80V @ A0.
ANALOG GROUND: on the PCB, R2 and C1 return straight to the solid B.Cu GND plane (offset is microvolts). Wired builds: return them to the ADS1115 GND pin, meeting power GND at ONE point (bench: 15mV rail-to-rail drop = -86mV error at the battery after the 5.7x scale-up).

Layout mirrors physical build order left to right: battery -> fuse -> protection/divider/buck -> ESP32-C3 hub -> ADS1115 / DHT22. PROT and GND are drawn as buses (top/bottom rails), like the two power rails on a breadboard -- every part along the rail solders to that same bus wire on the perfboard.